Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 104402

Subject: Analog Communication

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are **NINE** questions in this paper.
(iii) Attempt **FIVE** questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):-** **[2 x 7 = 14]**

(a) If the carrier of a 100% modulated A.M. wave is suppressed, the power saving will be in percent-
(i) 50
(ii) 75
(iii) 67
(iv) 80

(b) Modulation is the process of-
(i) combining audio and radio frequency waves
(ii) reducing distortion in RF amplifiers
(iii) generating constant-frequency radio waves
(iv) improving thermal stability of a transistor

(c) The FM broadcast band is from-
(i) 88 to 108 MHz
(ii) 88 to 108 kHz
(iii) 1600 kHz to 59.75 MHz
(iv) 88 to 108 GHz

(d) The standard intermediate frequency for FM radio is-
(i) 10.7 MHz
(ii) 10.7 kHz
(iii) 4.5 MHz
(iv) 4.5 kHz

(e) With 100% modulation, ratio of sideband power to total power transmitted in an amplitude-modulated wave is-
(i) 2/3
(ii) 4/3
(iii) 1/2
(iv) 1/3

(f) In an FM signal, as the modulation index increases, the power-
(i) increases
(ii) remains constant
(iii) decreases
(iv) None of above

(g) The simplex system is a _____ system
(i) bidirectional communication
(ii) multidirectional communication
(iii) unidirectional communication
(iv) None of the above

(h) In _____ modulation the phase of the carrier signal is modulated by the message signal.
(i) pulse
(ii) amplitude
(iii) phase
(iv) frequency

(i) In baseband signal the audio frequency signal range is around _____
(i) 10-20 kHz
(ii) 5-10 kHz
(iii) 0-20 kHz
(iv) 0-20 MHz

(j) In super heterodyne receiver, at high frequency the _____
(i) signal processing should be done
(ii) signal transmission should be done
(iii) bandwidth is low
(iv) Both (i) and (ii)

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Q.2 (a) Draw the block diagram of communication system. Explain the concept of frequency mixing [7]

(b) Explain the function of each block of communication system. [7]

Q.3 (a) Explain generation of DSB-SC signal with the help of balanced modulator using diodes. [7]

(b) Draw the frequency spectrum of DSB-SC modulation with necessary mathematical expressions. [7]

Q.4 (a) Write short note on Pre-Emphasis and De-Emphasis circuits. [7]

(b) Draw block diagram of Super-heterodyne AM receiver and explain function of each block. [7]

Q.5 (a) (i) The maximum peak-to-peak voltage of an AM wave is 12 mV and the minimum voltage is 4mV. Calculate the modulation factor. [7]

(ii) An audio signal of 1 KHz is used to modulate a carrier 500 KHZ. Determine side band frequencies and band width required.

(b) Describe thermal noise and shot noise. [7]

Q.6 (a) With block diagram explain the generation of PWM signals. [7]

(b) What are the differences between PAM, PWM, and PPM? [7]

Q.7 (a) The output signal from an AM modulator is: [7]

$s(t) = 5\cos(1800\pi t) + 20\cos(2000\pi t) + 5\cos(2200\pi t)$

(a) Determine the modulation index.

(b) Determine the ratio of the power in the sidebands to the power in the carrier.

(b) Explain (i) Signal to Noise Ratio (ii) Figure of merit. [7]

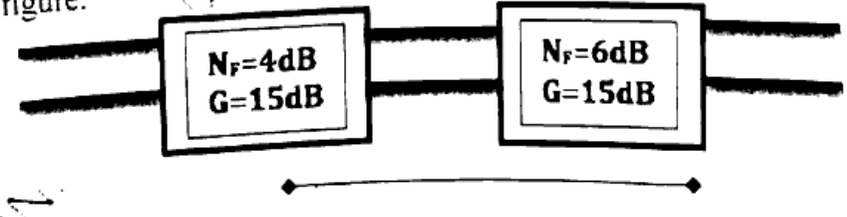
Discuss about noise effect in PM and obtain expression for figure of merit.

Q.8 (a) Define amplitude modulation and describe it in detail with suitable diagrams. [7]

(b) Write the difference forms of Amplitude Modulation, what is the efficiency of the AM? [7]

Q.9 (a) Write short notes on Envelope Detector. [7]

(b) Find the overall noise factor for the cascaded noise networks as shown in given figure. [7]



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